

1. Differentiate between ethics, law, and professional responsibility with suitable examples from the computing field.

Answer: Difference between Ethics, Law and Professional Responsibility (with computing Examples):

Ethics: Ethics refers to moral principles that guide what is right or wrong behavior. It is based on personal and societal values and is not legally enforced.

Example: A software developer chooses not to misuse user data even when there is no law preventing access to it.

Law:

Law consists of formal rules created and enforced by governments. Violating laws leads to legal punishment such as fines or imprisonment.

Example: Writing malware or performing unauthorized access (hacking) violates cybercrime laws and is legally punishable.

Professional Responsibility :

Professional responsibility includes the duties and standards expected from professionals in a specific field, often defined by professional bodies. It ensures competence, honesty, and accountability.

Example: Following the ACM code of Ethics by testing software properly before release to avoid harm to users.

In summary : Ethics is about moral choice, law is about legal obligation, and professional responsibility is about standards of conduct within the computing profession.

2. Discuss any two historical cases of ethical failures in technology and explain the lessons learned from them.

Answer: Two historical cases of ethical failures in technology and lessons learned:

1. Facebook - Cambridge Analytica Data Scandal

Facebook allowed third-party apps to collect personal data of users without proper consent, which was later used by Cambridge Analytica for political profiling and election influence.

Ethical failure:

Violation of user privacy, lack of informed consent, and poor data protection.

Lesson learned:

Technology companies must ensure data privacy, transparency, and user consent. Personal data should never be misused, even if it is legally accessible.

2. Therac-25 Software Failure

Therac-25 was a medical radiation machine whose software bugs caused patients to receive massive radiation overdoses, leading to serious injuries and deaths.

Ethical failure:

Inadequate software testing, ignoring error reports, and overreliance on software without safety checks.

Lesson learned:

In safety-critical systems, thorough testing, proper documentation, and fail-safe design are ethical responsibilities of computing professionals.

3. Explain the principles of software engineering ethics, highlighting the issues related to professional responsibility. Discuss how the ACM/IEEE code of Ethics guides ethical decision-making in software engineering practices.

Answer: Principles of Software Engineering Ethics and Professional Responsibility:

Software engineering ethics focuses on ensuring that software is developed and used in a way that benefits society and avoids harm. Key principles include honesty, integrity, competence, accountability, and respect for privacy. Software engineers must produce reliable software, disclose limitations, avoid conflicts of interest, and protect user data.

Issues related to professional responsibility include delivering low-quality or unsafe software, ignoring security vulnerabilities, misuse of confidential information, and failing to inform stakeholders

about risks. Professionals are responsible not only to employers but also to users, clients and society.

Role of the ACM / IEEE Code of Ethics :

The ACM / IEEE code of Ethics provides clear guidelines to support ethical decision-making in software engineering. It emphasizes:

- Acting in the public interest
- Maintaining professional competence
- Being honest and transparent in work.
- Respecting privacy, confidentiality, and intellectual property.

4. Compare and contrast Utilitarianism and Deontology (Kantian ethics) as ethical frameworks. How would each approach handle a software privacy dilemma?

Answer: Comparison of Utilitarianism and Deontology (Kantian ethics):

Utilitarianism:

Utilitarianism judges an action by its consequences. An action is ethical if it produces the greatest good for the greatest number of people. The outcome matters more than rules or intentions.

Deontology (Kantian Ethics):

Deontology, proposed by Immanuel Kant, judges an action based on duty, rules, and moral principles, not outcomes. An action is ethical if it respects moral duties and treats individuals as ends, not as means.

Handling a Software Privacy Dilemma:

Scenario: A company considers sharing users data to improve public services.

• Utilitarian approach:

sharing data may be considered ethical if it benefits a large number of people, even if some users' privacy is compromised.

• Deontological (Kantian) approach:

sharing data without consent is unethical because it violates the duty to respect user privacy, regardless of the benefits.

5. Explain Virtue Ethics and describe how it can guide ethical decision-making for software engineers.

Answer: Virtue Ethics and its role in Software engineering:

Virtue Ethics is an ethical framework that focuses on the character and moral virtues of a person rather than rules or consequences. It emphasizes developing good qualities such as honesty, responsibility, fairness, integrity, and empathy. The idea originates from Aristotle, who argued that ethical behavior comes from being a good person.

Guidance for software engineers:

Virtue Ethics guides software engineers to ask, "What would a good and responsible engineer do?" Instead of only following rules, engineers are encouraged to:

- Write reliable and secure code.
- Be honest about software limitations and risks.
- Respect user privacy and data
- Take responsibility for the social impact of their software.

6. Why is explainability a key ethical requirement for AI systems used in high-stakes domains like credit lending or criminal justice? Discuss two societal harms that can occur if an AI system operates as a black box.

Answer: Explainability as a key ethical requirement in high-stakes AI systems:

Explainability means that an AI system's decisions can be understood, interpreted, and justified by humans. In high-stakes domains like credit lending and criminal justice, AI decisions directly affect people's rights, freedom, and economic opportunities.

Ethical use requires that affected individuals and authorities can understand why a decision was made, challenge it if necessary, and ensure it is fair and lawful.

Social

Societal harms of black-box AI systems:

1. Unfair discrimination and bias :

If an AI system operates as a black box, hidden biases in data or algorithms may lead to discriminatory outcomes. Without explainability such as bias cannot be detected or corrected.

2. Lack of accountability and due process :

When decisions cannot be explained, it becomes difficult to assign responsibility or allow individuals to appeal decisions. This can undermine justice, transparency, and trust in institutions using AI.

7. What are the key principles of the ACM Code of Ethics? Why are such professional codes important for computing professionals?

Answer: The ACM code of Ethics provides ethical guidelines for computing professionals. Its key principles include:

- Contribute to society and human well-being by ensuring technology benefits people.
- Avoid harm such as privacy violations, security risks, or unsafe systems.
- Be honest and trustworthy in professional work.
- Respect privacy and confidentiality of users' data.
- Maintain professional competence and follow quality standards.
- Accept responsibility and accountability for one's work and its consequences.

Importance of professional codes for computing professionals:

Professional codes are important because they:

- provide clear ethical guidance for decision-making in complex situations.
- Promote public trust in computing technologies and professionals.
- Help professionals balance duties to employers, users, and society.
- Encourage responsible, fair and lawful conduct in software development and use.

8. What is whistleblowing? Discuss the ethical considerations and challenges a computing professional might face when deciding to blow the whistle.

Answer: Whistleblowing is the act of reporting illegal, unethical, or harmful practices within an organization to authorities or the public by an employee or professional.

Ethical considerations for a computing professional:

• Public interest and harm prevention:

A computing professional has an ethical duty to protect users and society if software systems cause serious harm, such as data breaches, unsafe systems, or misuse of data.

• Professional responsibility and loyalty:

Professionals face a conflict between loyalty to the employer and responsibility to society. Ethical codes prioritize public welfare over organizational

secrecy.

challenges faced when whistleblowing:

• Risk of retaliation:

The individual may face job loss, legal action, or professional isolation.

• Uncertainty and evidence:

It can be difficult to prove wrongdoing or decide whether the issue is serious enough to justify whistleblowing.

9. The ACM code of Ethics emphasizes both contribute to society and human well-being and respect privacy. Describe a realistic scenario where these two principles might come into conflict for a data scientist. How might the code guide a professional in resolving this tension?

Answer: Conflict between "contribute to society" and "respect privacy":

Scenario:

A data scientist working for a public health organization analyzes large-scale mobile location and health data to predict the spread of an infectious disease. Using detailed personal data could significantly help society by improving early warnings and saving lives. However, this data contains sensitive personal information, and analyzing it may violate individuals' privacy.

Here, the principle of contributing to society and human well-being conflicts with the principle of respecting privacy.

How the ACM code of Ethics guides resolution:

The ACM code of Ethics guides the professional to:

- Minimize harm by using the least intrusive data necessary.
- Apply anonymization and aggregation to protect individual identities.
- Ensure transparency and informed consent where possible.
- Use data only for the stated public benefit and prevent misuse.

10. Explain the differences between copyright, patents, and trademarks in the context of software and digital products.

Answer:

1. Copyright:

- Protects the expression of ideas, such as source code, documentation, or digital media.
- Automatically exists when the work is created; no registration is strictly needed.
- Example: A developer's unique code for a mobile app is copyrighted; others cannot copy it without permission.

2. Patent

- Protects novel inventions or processes, including software algorithms or technical solutions that are new, non-obvious, and useful.
- Requires formal registration with a patent office and has a limited duration (usually 20 years).

- Example: A new encryption algorithm implemented in software can be patented to prevent others from using the same method.

3. Trademark

- Protects brand names, logos, or symbols that identify the source of a product or service.
- Prevents confusion in the marketplace but does not protect the underlying software code or ideas.
- Example: The logo of a software company or the name of a digital product, like 'Photoshop', is trademarked.

11. compare open-source licensing with proprietary software licensing. what are the ethical implications of each model?

Feature	Open-source Licensing	Proprietary Software Licensing
Source Code	Freely accessible	Restricted; hidden from users.
Modification	Allowed; users can modify and improve software	Not allowed; modifications prohibited
Distribution	can be freely shared and redistributed	controlled; redistribution requires permission.
cost	often free	Usually paid; commercial
Ethical Focus	Promotes collaboration, transparency, and knowledge sharing.	Emphasizes respect for intellectual property and legal compliance
Responsibility	credit original authors follow license terms.	Avoid piracy; comply with licensing agreements
Example	Linux, Apache Server	Microsoft Windows, Adobe Photoshop

12. Define professional dissent whistleblowing in computing. What are the key ethical considerations and potential risks a professional must weigh before deciding to blow the whistle on harmful or unethical practices within their organization?

Answer: Professional Dissent:

Professional dissent occurs when a computing professional disagrees with or challenges decisions, policies, or practices within their organization that they believe are unethical, unsafe, or harmful, while remaining part of the organization.

Whistleblowing:

Whistleblowing is the act of publicly or formally reporting unethical, illegal, or harmful practices within an organization to external authorities, regulators, or the public when internal dissent fails.

Key Ethical Considerations:

1. Public Interest: The professional has a duty to protect users, society, or stakeholders from harm caused by software or digital systems.
2. Professional Responsibility: Ethical codes require honesty, integrity, and accountability.
3. Proportionality: Assess whether the harm prevented justifies the potential consequences of whistleblowing.
4. Due Process: Attempt internal resolution first, if feasible, before escalating externally.

Potential Risks:

- Retaliation: Job loss, demotion or legal threats from the employer.
- Professional isolation: Damage to reputation or strained relationships with colleagues.
- Emotional and financial stresses: Whistleblowing can be personally taxing and risky.

13. Discuss the concept of informed consent in personal data collection. What ethical principles should organisations follow when collecting user data?

Answer: Concept:

Informed consent means that individuals are fully aware of what personal data is being collected, how it will be used, and any potential risks, and they agree voluntarily to provide it. Consent must be clear, specific and revocable at any time.

Ethical principles organizations should follow:

1. Transparency: Clearly explain what data is collected, why, and how it will be used.

2. Privacy: Collect only data that is necessary and store it securely to prevent unauthorized access.

3. Voluntariness: Users must have a genuine choice to give or withhold consent without coercion.

4. Accountability: Organizations must take responsibility for handling data ethically and in compliance with relevant laws.

5. Respect for rights: Allow users to access, correct, or delete their data when requested.

14. What are the ethical concerns associated with surveillance technologies? Explain with examples how these technologies can be misused.

Answer: Surveillance technologies, such as CCTV cameras, facial recognition systems, and online tracking tools, raise several ethical concerns because they monitor individuals' activities and collect sensitive data.

Key Ethical concerns:

1. Privacy Violation: Continuous monitoring can intrude on personal privacy.

Example: Using facial recognition in public spaces without consent can track people's movements and identify individuals without their knowledge.

2. Consent and Transparency:

Individuals are often unaware of how their data is collected or used.

Example: Mobile apps that track location or behavior without clearly informing users.

3. Discrimination and Bias:

Algorithms in surveillance systems can reflect biases, leading to unfair targetting.

Example: Facial recognition systems misidentifying people of certain ethnic groups more frequently, resulting in wrongful suspicion or arrests.

4. Abuse of Power:

Authorities or organizations can misuse surveillance to control, intimidate, or manipulate populations.

Example: Governments monitoring political activists or employees being excessively monitored to enforce strict workplace control.

16. What is algorithmic bias ? Explain its causes and suggest measures to ensure fairness in AI systems.

Answer:

Algorithmic Bias:

Algorithmic bias occurs when AI systems produce unfair or discriminatory outcomes against certain groups, often reflecting biases in data or design.

Causes:

- Biased training data: Historical inequalities reflected in datasets.
- Feature selection: Using variables that unintentionally discriminate.
- Lack of diversity: Limited perspectives in development teams.

• Opaque models: Difficulty in detecting bias in complex algorithms.

Measures to Ensure Fairness :

- Use diverse and representative data.
- Conduct regular bias testing and audits.
- Promote transparency and explainable AI.
- Involve inclusive teams and follow ethical guidelines.

17. Discuss the importance of transparency and explainability in AI. Who should be held accountable when an autonomous system causes harm?

Answer: Transparency and Explainability in AI:

Transparency means that an AI system's functions, data, and decision-making processes are clear and understandable.

Explainability ensures that humans can interpret and justify the system's decisions.

These principles are crucial in AI applications affecting healthcare, finance, or criminal justice, where decisions can significantly impact people's lives.

Transparency builds trust, accountability, and fairness, and helps detect errors or bias.

Accountability for Harm :

When an autonomous system causes harm, responsibility can involve multiple parties;

- Developers : For designing, testing, and maintaining the AI responsibly.

- Organizations : For ensuring proper use, oversight and monitoring,

- Regulators : For enforcing ethical and legal standards.

Ethical frameworks and professional codes emphasize that humans must remain accountable, even when systems act autonomously.

18. Discuss the role of ethical leadership in IT organizations. How can leaders foster an ethical culture within their teams?

Answer: Ethical leadership in IT involves guiding teams with integrity, fairness and responsibility, ensuring that technology is developed and used for the benefit of society.

Fostering an Ethical Culture:

1. Lead by Example: Demonstrate honesty, accountability, and respect in all decisions.

2. Establish clear Guidelines: Promote adherence to ethical codes and organizational policies.

3. Encourage Open Communication: Allow team members to voice concerns, raise ethical issues, or dissent without fear of retaliation.

4. Provide Training and Awareness: Educate staff on ethical practices, data privacy and responsible AI use.

5. Reward Ethical Behavior: Recognize and reinforce actions that uphold integrity and social responsibility.

19. Differentiate between ethical hacking and cybercrime.
Under what conditions is hacking considered ethical?

Answer: Ethical Hacking vs Cybercrime:

Aspect	Ethical Hacking	Cybercrime
Definition	Authorized attempts to find and fix security vulnerabilities	Unauthorized access or attacks to steal, damage, or disrupt systems.
Permission	Always conducted with consent from the system owner.	Illegal; performed without consent.
Purpose	Improve security, protect data, prevent attacks.	Personal gain, sabotage, theft, or harm.
Legality	Legal and recognized professionally	Illegal; punishable under law
Example	Penetration testing for a company's network.	Hacking a bank account or deploying ransomware.

Conditions for Hacking to be Ethical:

1. Explicit authorization from system owners,
2. clear objectives to identify and fix vulnerabilities
3. Non-malicious intent; no harm or theft caused.
4. Confidentiality and responsible reporting of findings.

20. Explain the concept of responsible disclosure in cybersecurity. What ethical guidelines should a security researcher follow when discovering a vulnerability?

Answer: Responsible Disclosure in cybersecurity:
Concept: Responsible disclosure is the practice where a security researcher privately reports a discovered vulnerability to the affected organization and allows them adequate time to fix the issue before making it public.

Ethical guidelines for a security researcher:

- Act in good faith: Discover vulnerabilities without malicious intent.
- Notify the organization first: Share full details confidentially with the vendor or system owner.

- Allow reasonable time for remediation: Give the organization sufficient time to patch the vulnerability.
- Avoid exploitation: Do not misuse, sell, or publicly expose the vulnerability prematurely.
- Respect laws and professional codes: Follow legal boundaries and ethical standards (e.g. the ACM code of Ethics).

21. What is the digital divide? Discuss its social implications and suggest measures to bridge this gap.!

Answer: Digital Divide:

The digital divide refers to the gap between individuals or groups who have access to digital technologies and those who do not.

Social implications:

- Educational inequality: Students without internet access fall behind in online learning.
- Economic disadvantage: Limited access reduces job opportunities and digital skills.
- Social exclusion: People may be excluded from online services, e-governance, and information.

Measures to bridge the gap:

- Expand affordable internet infrastructure, especially in rural areas.
- Provide low-cost devices and public access centers.
- Promote digital literacy and training programs.
- Support inclusive government and private initiatives.

Reducing the digital divide is essential for social equity, economic development, and inclusive growth.

22. Explain the concept of green computing. How can computing professionals contribute to environmental sustainability?

Answer: Green Computing:

Green computing refers to the environmentally responsible use of computers and IT systems by minimizing energy consumption, reducing e-waste, and lowering the carbon footprint throughout a system's lifecycle.

Role of computing professionals:

- Design energy-efficient software and algorithms that reduce processing and power use.
- Promote efficient hardware usage, virtualization and cloud optimization.

- Encourage responsible e-waste management, recycling and reuse of devices.

- Support sustainable IT policies, such as using renewable energy-powered data centers.

conclusion: Through conscious design and responsible practices, computing professionals can significantly contribute to environmental sustainability and climate protection.

23. What is a conflict of interest in the IT workplace? Provide examples and explain how professionals should ethically handle such situations.

Answer: Conflict of interest in the IT Workplace:

A conflict of interest occurs when an IT professional's personal, financial, or other interests interfere, or appear to interfere, with their professional duties and objectivity.

Examples:

- A software manager awarding a contract to a company owned by a family member.
- A developer reviewing or approving software in which they hold financial shares.
- An IT consultant promoting a tool for personal

gain rather than technical merit.

Ethical handling of conflicts:

- Disclose the conflict honestly to employers or clients.
- Recuse oneself from decision-making where bias may exist.
- Follow organizational policies and professional codes.
- Prioritize fairness, transparency, and public interest.

Ethical managing conflicts of interest preserves trust, integrity, and professionalism in the IT workplace.

24. Explain the concept of justice and fairness according to any one religious tradition or ideology, such as Islam, Hinduism, Christianity, Buddhism, Judaism or Communism. How can these principles be applied to promote ethical decision-making in modern professional life?

Answer: Justice and Fairness in Islam and their application in professional life:

Concept in Islam:

In Islam, justice and fairness are core moral principles. Justice means treating all people equally, honestly, and without bias, regardless of status, wealth, or power. Islam strongly condemns oppression, corruption, and dishonesty, and emphasizes accountability, truthfulness, and respect for others' rights in all actions.

Application in modern professional life:

- Fair decision-making: Professionals should avoid favoritism and discrimination in hiring, promotion, and evaluation.
- Honesty and integrity: Truthful reporting, transparent communication, and avoidance of fraud or deception.
- Accountability: Taking responsibility for mistakes and ensuring work does not harm others.
- Respect for rights: Protecting privacy, dignity, and lawful interests of clients, users, and colleagues.

25. What is Aristotle's concept of the Golden Mean, and how does it guide individuals to find a balanced and virtuous course of action between extremes in ethical decision making?

Answer:

Aristotle's Concept of the Golden Mean:

The Golden Mean is an ethical principle proposed by Aristotle, which states that virtue lies between two extremes - excess and deficiency.

Ethical behavior is achieved by finding a balanced, moderate course of action appropriate to the situation.

Guidance in ethical decision-making:

According to Aristotle, each virtue is a mean between two vices. For example, courage lies between recklessness and cowardice. Individuals should use practical wisdom to judge the right balance rather than rigid rules.

The Golden Mean guides individuals to act virtuously by avoiding extremes and choosing moderation, balance, and rational judgement in ethical decisions.

28. Discuss the issue of moral decay in Bangladesh by analysing its major causes, social and economic effects, and possible strategies to address and prevent it.

Answer: Major causes:

Moral decay in Bangladesh is driven by corruption, lack of ethical education, weak rule of law, unemployment, and misuse of power, social media misuse, declining family supervision, and erosion of religious and moral values also contribute to unethical behavior in society.

Social and economic effects:

Socially, moral decay leads to loss of trust, crime, dishonesty, and social injustice. People

become less respectful of laws and others' rights. Economically, corruption and unethical practices discourage investment, reduce productivity, and widen inequality, slowing national development.

Strategies to address and prevent moral decay:

- Strengthen ethical and moral education from an early age.
- Ensure strict enforcement of laws and accountability.
- Promote ethical leadership in politics, business and institutions
- Create employment opportunities to reduce frustration and crime.
- Encourage family, religious, and social institutions to instill values.